

6 There are some counters in a bag.  
The counters are red or white or blue or yellow.

Bob is going to take at random a counter from the bag.

The table shows each of the probabilities that the counter will be blue or will be yellow.

|                    |     |       |      |        |
|--------------------|-----|-------|------|--------|
| <b>Colour</b>      | red | white | blue | yellow |
| <b>Probability</b> |     |       | 0.45 | 0.25   |

There are 18 blue counters in the bag.

The probability that the counter Bob takes will be red is twice the probability that the counter will be white.

(a) Work out the number of red counters in the bag.

.....  
(4)

A marble is going to be taken at random from a box of marbles.  
The probability that the marble will be silver is 0.5

There must be an even number of marbles in the box.

(b) Explain why.

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.....  
(1)

**(Total for Question 6 is 5 marks)**